

Visualization of flight routes

Packages needed

```
library(tidyverse)
library(usmap)
library(sf)
```

Dataset

The `flight_df_raw` data we have here is a complete record of all domestic flights for American, Delta, United, and Southwest in 2019 (`nycflights13::flights` contains only flights departing from NYC).

```
flight_df_raw <- read_csv("data/flight_df.csv") |> select(-DepTime, -ArrTime)
airports_raw <- read_csv("data/airports.csv")
us_map_sf <- us_map(regions = 'states') |> filter(!abbr %in% c("AK", "HI"))
```

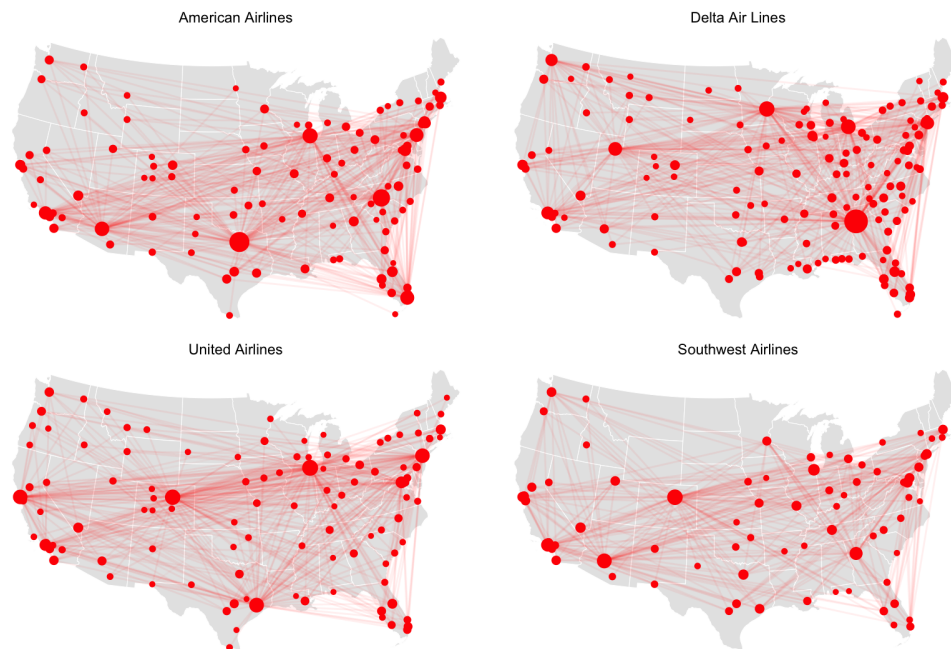
This is some wrangling to match the coordinate reference system (CRS) of the airports with the CRS of the US map (week 6 Monday).

```
airports <- airports_raw |>
  # create an sf object
  st_as_sf(coords = c("x", "y"), crs = 4326) |>
  # transform to the same CRS as the us_map_sf
  st_transform(crs = st_crs(us_map_sf)) |>
  # extract the coordinates
  mutate(x = st_coordinates(geometry)[,1], y = st_coordinates(geometry)[,2]) |>
  # focus on airports on continental US
  filter(between(x, -3000000, 3000000), y < 700000) |>
  # clean up
  rename(airport = ident) |>
  as_tibble() |>
  select(airport, x, y)
```

airports

```
## # A tibble: 19,317 x 3
##   airport      x      y
##   <chr>    <dbl>  <dbl>
## 1 OJ0      1399199. -1366000.
## 2 2A8      1180494. -1107747.
## 3 AL03     1149352. -1265974.
## 4 EET      1230576. -1216678.
## 5 AL39     1226771. -1208917.
## 6 8A0      1262389. -1093420.
## 7 ALX      1310365. -1233325.
## 8 AIV      1100991. -1243840.
## 9 AL67     1138225. -1537664.
## 10 EKY     1215351. -1203918.
## # i 19,307 more rows
```

Your goal



How many layers (`geom_*()`) do you think this plot contains? Sketch a general `ggplot2` skeleton you could use to build it.

Your answer: write your syntax

Points representing the airports

What should your airport point data look like to fit the `ggplot2` syntax above? Sketch the data frame you'd want: What does each row represent? Which columns does it need?

Your answer:

Can you implement the above from the `flight_df_raw` data? You may iterate a few times between these two steps.

```
nodes_df <- flight_df_raw |>
  # this function standardizes the variable names to lowercase with underscores
  janitor::clean_names()
```

Lines representing the routes

Repeat the same process for the routes. What should your airport point data look like to fit the `ggplot2` syntax above? Again what does each row represent, and which columns does it need?

Your answer:

Now build this route data frame from `flight_df_raw`. Again, you may need to iterate between sketching and coding.

```
routes_df <- flight_df_raw |>
  janitor::clean_names()
```

Final plot - Assembling the components

With a data frame ready for each layer, put everything together and build the final plot.

When you run your code, `ggplot2` may produce warnings like these:

- 1: Removed xxx rows containing missing values or values outside the scale range (``geom_point()``).
- 2: Removed xxx rows containing missing values or values outside the scale range (``geom_line()``).

These warnings tell you that some rows have missing values, so `ggplot2` dropped them before plotting. Can you investigate into these missingness?